

WEEKLY DEV 3

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CSC426

ANSWER

1. Link to the questionnaire:

https://docs.google.com/forms/d/e/1FAIpQLSfV2S4KP_PhulYlgdVZ67kKhdf8JRQHE8Ekg_tQ0uJnzWdBpBA/viewform?usp=header

2. Analysis of the issue with Ishikawa problem analysis tool

EFFECT: Ineffective Reporting and Resolution of Community & Student Issues

Main Cause Categories

1. People: Possible causes:

- Low awareness of reporting channels
- Fear of victimization
- Lack of trust in authorities
- Poor cooperation among stakeholders

2. Process: Possible causes:

- No standardized reporting procedure
- Slow response from authorities
- Manual complaint handling
- No tracking of reported issues

3. Technology: Possible causes:

- No centralized reporting platform
- Limited internet access
- Poor system availability
- Lack of mobile-friendly reporting

4. Management: Possible causes:

- Poor coordination among agencies
- Lack of accountability
- Delayed decision making
- Insufficient monitoring

5. Environment: Possible causes:

- Poor road conditions
- Insecurity
- Poor lighting
- Waste disposal challenges
- Flood-prone areas

6. Policy / Resources: Possible causes:

- Inadequate funding
- Weak enforcement of regulations
- Lack of reporting policies
- Insufficient personnel
- Limited public awareness campaigns

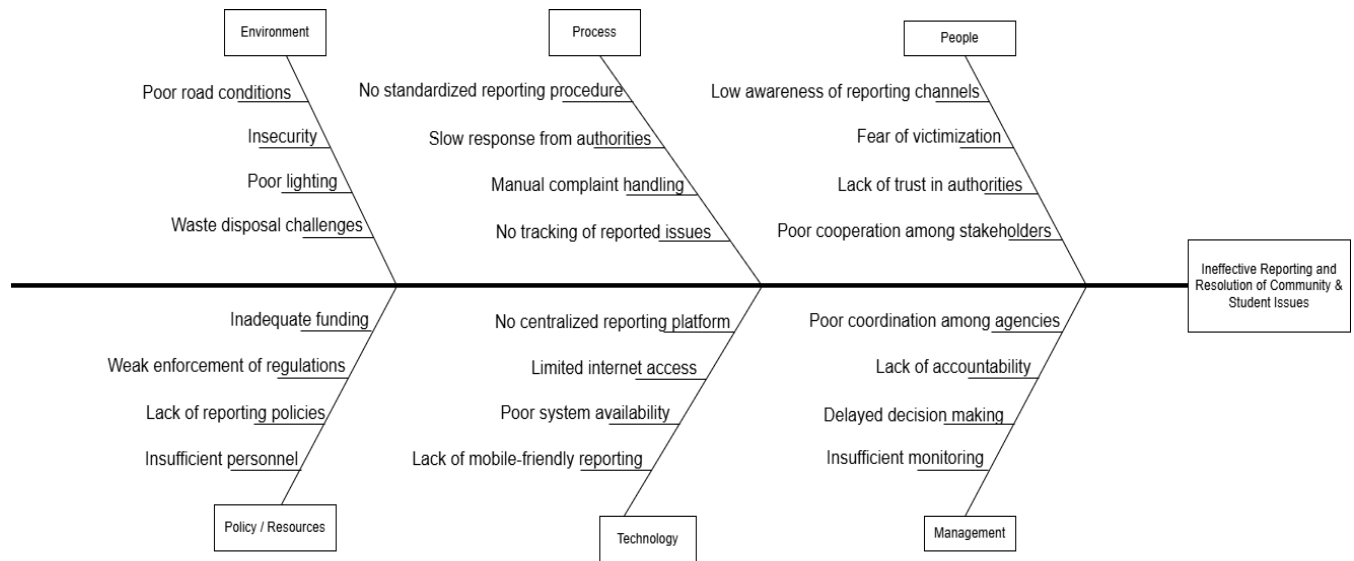


Figure 1: ISHIKAWA DIAGRAM

3. Sketch use-case diagram(s) to show how your system/app interacts

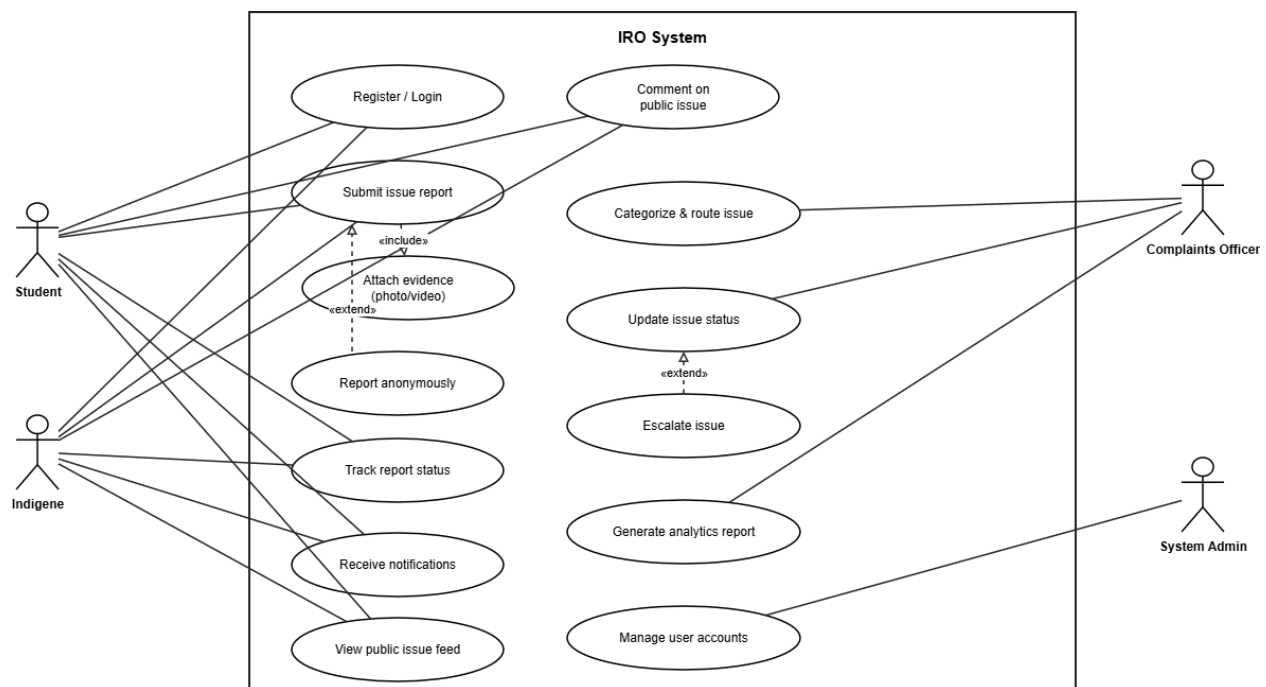


Figure 2: USE CASE DIAGRAM

Software Requirements Specification (SRS)

IRO (Issues Reporting Outlet)

Version: 1.0

Project Type: Mini Web-Based Reporting System

Department: Computer Science

Institution: Adekunle Ajasin University, Akungba-Akoko

Introduction

Purpose

The Issues Reporting Outlet (IRO) is a web-based reporting platform developed to provide AAUA students and members of the host communities (Akungba, Iwaro, Etioro, Ayegunle and surrounding areas) with a centralized medium for reporting community-related issues.

The system allows users to submit complaints, receive a unique Complaint ID, and monitor the progress of their reports through a tracking interface.

Scope

The IRO system enables users to:

1. Submit complaints
2. Attach evidence (optional)
3. Report anonymously
4. Generate Complaint IDs
5. Store reports locally
6. Track submitted complaints
7. View complaint status

8. Read information about the project

The current prototype operates entirely on the client side using Local Storage and does not require an internet server or database.

Intended Users

The application is designed for:

1. AAUA Students
2. Community Indigenes
3. Student/Community Members
4. University Authorities
5. Community Leaders

Overall Description

System Perspective

IRO is a standalone web application consisting of four major interfaces:

- a) Home Page
- b) Report Issue Page
- c) Track Report Page
- d) About Page

JavaScript provides the interaction logic while Local Storage temporarily stores submitted complaints.

Product Functions

The system performs the following functions:

1. Accept issue reports
2. Generate unique Complaint IDs
3. Store reports
4. Validate user inputs
5. Allow anonymous reporting
6. Search reports
7. Display report status
8. Provide project information

System Components and Interaction

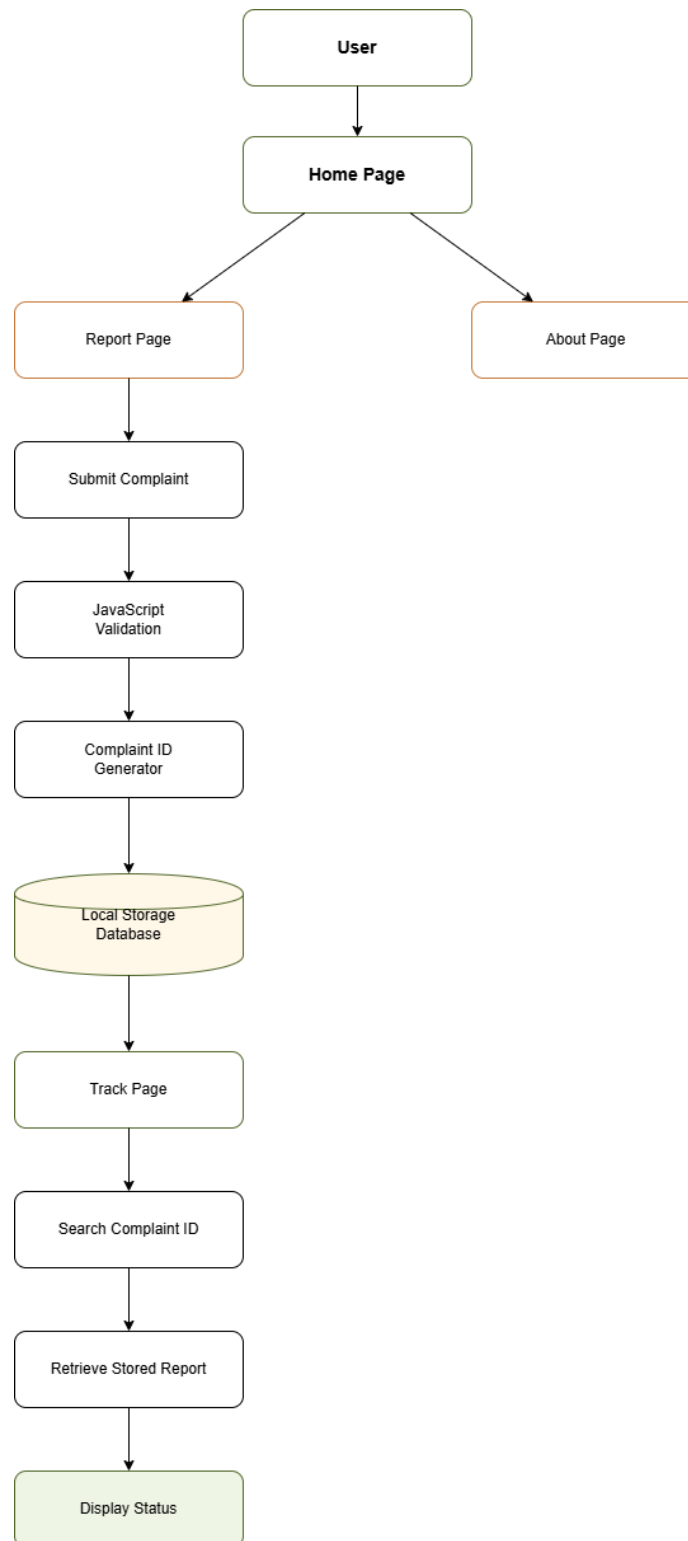


Figure 3: System Components and Interaction

Functional Requirements

1. **FR1 – Navigation:** The system provides navigation between all pages.
2. **FR2 – Submit Report:** The system allows users to submit a complaint containing:
 - a) Name
 - b) Email
 - c) Phone Number
 - d) User Category
 - e) Community
 - f) Issue Category
 - g) Location
 - h) Title
 - i) Description
 - j) Optional Image
3. **FR3 – Anonymous Reporting:** The system allows users to submit reports anonymously.
4. **FR4 – Input Validation:** The system verifies that required fields are completed before submission.
5. **FR5 – Complaint ID Generation:** The system automatically generate a unique Complaint ID after successful submission. Example: IRO-20260710-4827
6. **FR6 – Local Storage:** The system saves submitted reports inside the browser using Local Storage.
7. **FR7 – Complaint Tracking:** The system allows users to search using their Complaint ID.
8. **FR8 – Display Complaint Information:** After a successful search, the system displays:
 - a) Complaint ID
 - b) Title

- c) Description
- d) Category
- e) Community
- f) Location
- g) Status
- h) Date Submitted
- i) Current Stage

9. **FR9 – Invalid Search Handling:** If no complaint exists, the system displays an appropriate error message.

10. **FR10 – Responsive Design:** The system automatically adjusts to:

- a) Desktop
- b) Tablet
- c) Mobile phones

Non-Functional Requirements

1. Performance:

- a) Pages should load within 2 seconds.
- b) Report submission should complete instantly.
- c) Complaint retrieval should occur immediately.

2. Usability: The interface:

- a) Is simple to understand.
- b) Uses readable fonts.
- c) Uses clear navigation.
- d) Require minimal technical knowledge.

3. Reliability: The application:

- a) Store reports correctly.
- b) Prevent incomplete submissions.
- c) Maintain complaint information during browser sessions unless Local Storage is cleared.

4. Security: The prototype:

- a) Validate required fields.
- b) Restrict accepted upload formats to images.
- c) Generate unique Complaint IDs.

Since this is a prototype, user authentication and encryption are not implemented.

5. Availability: The application shall be available whenever opened in a modern web browser.

6. Maintainability: The source code is separated into:

- I. HTML
- II. CSS
- III. JavaScript
- IV. SVG Assets

to simplify future maintenance.

7. Portability: The application runs on modern browsers including:

- I. Google Chrome
- II. Microsoft Edge
- III. Mozilla Firefox
- IV. Opera

without requiring installation.

User Interface Requirements

The interface contains:

1. Home Page

- a. Hero Section
- b. Features
- c. Categories
- d. FAQ
- e. Contact Section

2. Report Page

- a. Complaint Form
- b. Validation
- c. Anonymous Reporting
- d. Complaint ID Display

3. Track Page

- a. Complaint Search
- b. Status Display
- c. Progress Timeline

4. About Page

- a. Project Description
- b. Objectives
- c. Stakeholders
- d. Benefits

Data Requirements

Each complaint record contains:

Table 1: Data Requirements

Field	Description
Complaint ID	Unique identifier
Full Name	Reporter's name
Email	Contact email
Phone	Phone number
User Type	Student/Indigene
Community	Selected community
Category	Issue category
Location	Exact location
Title	Issue title
Description	Detailed complaint
Anonymous	Yes/No
Date Submitted	Submission date
Status	Pending / In Progress / Resolved

Assumptions

1. Users have access to a web browser.
2. JavaScript is enabled.
3. Browser Local Storage is available.
4. Users provide truthful information.

Constraints

1. No backend server.
2. No online database.
3. Reports are stored locally in the browser.
4. Uploaded images are not permanently stored.
5. Complaint status is simulated for demonstration purposes.

Future Enhancements

Future versions of IRO may include:

1. Administrator dashboard
2. User authentication
3. Cloud database integration
4. Email notifications
5. SMS notifications
6. GIS location mapping
7. Analytics dashboard
8. Complaint assignment to departments

9. Real-time status updates
10. Mobile application

Conclusion

The IRO system provides a centralized platform for reporting and monitoring issues affecting AAUA students and the surrounding communities. The prototype demonstrates the core functionality of an issue reporting system through complaint submission, Complaint ID generation, local data storage, and complaint tracking. The modular design and responsive interface provide a solid foundation for future enhancements, including server-side processing, database integration, and administrative management features.